

Day 18 of training

Mapping in React

Definition:

Mapping in React is the process of iterating over an array and displaying its elements in the UI. It is done using the `map()` method of JavaScript. Mapping is used to render lists dynamically instead of writing each element manually.

Key Points:

1. Mapping is mainly used to render arrays of data in JSX.
2. Every element in a mapped list should have a unique `key` prop to help React identify elements.
3. Can be used with arrays of strings, numbers, or objects.
4. `map()` returns a new array of JSX elements.
5. Helps in making UI dynamic, reusable, and maintainable.

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```
function FruitsList() {  
  
  const fruits = ["Apple", "Banana", "Mango", "Orange"];  
  
  return (  
  
    <ul>  
  
      {fruits.map((fruit, index) => (  
  
        <li key={index}>{fruit}</li>  
  
      ))}  
  
    </ul>  
  
  );  
  
}
```

- `fruits.map()` iterates over each fruit.

- `key={index or id }` is required for React to track list items efficiently.

Advantages of Mapping in React:

1. Renders dynamic lists easily.
2. Keeps code clean and maintainable.
3. Avoids repetition of HTML.
4. Works well with props and child components.
5. Essential for real-world apps with data from APIs.